

Installing R and RStudio

for the Third INFOGUT Training School

Bioinformatics 101 for colon model experiments

macOS

The key idea: R and RStudio are two different programs. Install R first, then install RStudio. R executes your code; RStudio is the interface you will normally use during the course.

Beginner-friendly guide based on official documentation from the R Project (CRAN) and Posit.

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1. Before you start

For this course you need both R and RStudio Desktop installed on your computer before the hands-on sessions.

Software	What it does
R	The programming language and statistical computing environment. It actually runs your R code.
RStudio Desktop	A free IDE that provides a convenient interface for writing, running, organizing, and viewing R code and results.

Recommended order:

1. Install the latest stable version of R.
2. Install the latest stable version of RStudio Desktop.
3. Open RStudio.
4. Run the verification tests at the end of this guide.

Important: Do not install only RStudio. RStudio requires R to be installed on the computer first.

2. Install R on macOS

Step 1 - Check whether your Mac uses Apple silicon or Intel

Open Apple menu → About This Mac and look at the Chip or Processor field.

- Apple silicon: the chip is Apple M1, M2, M3, M4, M5, etc.
- Intel: the processor field says Intel.

Step 2 - Open the official CRAN macOS page

CRAN for macOS: <https://cran.r-project.org/bin/macosx/>

Step 3 - Download the correct installer

- Apple silicon: choose the latest stable package ending in -arm64.pkg.
- Intel Mac: choose the latest stable package ending in -x86_64.pkg.

Important: Make sure you download the installer that matches your Mac processor. If you are unsure, check “About This Mac” before downloading.

Step 4 - Install R

5. Open the downloaded .pkg file.
6. Click Continue through the installer.
7. Accept the software licence when requested.
8. Keep the default installation location.
9. Click Install.
10. Enter your Mac password or use Touch ID if macOS asks for authorization.

11. When installation finishes successfully, close the installer.

Beginner note: You normally do not need XQuartz, Xcode, Fortran, or additional compilers just to start the course. Some advanced R packages may need extra tools later.

3. Install RStudio Desktop

Install RStudio only after R has been installed successfully. For the Training School, the free RStudio Desktop edition is sufficient.

Official Posit downloads: <https://posit.co/downloads/>

12. On the Posit downloads page, click the “Download RStudio” button.
13. You will be redirected to the official RStudio IDE User Guide, where a “Direct Downloads” table is displayed.
14. In the table, find the “macOS 14+” row and click the .dmg installer (for example, RStudio-2026.08.2-200.dmg).
15. When the download is complete, open the downloaded .dmg file.
16. Drag the RStudio icon into the Applications folder, then eject the RStudio disk image.
17. Open RStudio from Applications.

Apple silicon: If RStudio reports that Rosetta 2 is required on your Mac, follow the macOS prompt to install it.

Note: the exact RStudio version number may change. Always choose the latest stable version listed for your operating system.

4. Open RStudio and verify the installation

Open RStudio. It should automatically detect your R installation. In the standard layout, the Console is usually in the lower-left part of the window.

Test 1 - Basic calculation

```
1 + 1
```

Expected result:

```
[1] 2
```

Test 2 - Check the R version

```
R.version.string
```

Test 3 - Show session information

```
sessionInfo()
```

Success: If these commands run without an error, R and RStudio are correctly connected and your computer is ready for the course.

5. Create and run your first R script

18. In RStudio, choose File → New File → R Script.

19. Type the following code in the script editor:

```
x <- 5  
y <- 10
```

```
x + y
```

Run the current line or selected code using:

- Cmd + Enter

You should see the result in the Console:

```
[1] 15
```

Good habit: Write and save your work in R scripts rather than typing everything directly into the Console. This makes your analysis easier to repeat and share.

6. Optional test: install an R package

R includes many functions by default. Additional functionality is distributed through packages. To test package installation, you can install ggplot2:

```
install.packages("ggplot2")
```

You normally install a package only once. To use it in a new R session, load it with:

```
library(ggplot2)
```

7. Common beginner problems

Problem	What to try
RStudio opens but cannot find R	Confirm that R was installed before RStudio, then close and reopen RStudio.
I installed R but cannot find RStudio	R and RStudio are separate programs. Download RStudio Desktop from the official Posit website.
I see several R downloads	Choose the latest stable/base release for your operating system, not a development build.
A package will not install	Read the complete error message. Some packages require additional system libraries or compilers; this is separate from the basic installation.
macOS says the application cannot be opened	Make sure you downloaded the current R/RStudio build compatible with your macOS version and processor architecture.

8. Final checklist before the Training School

- ☐ R is installed.
- ☐ RStudio Desktop is installed.
- ☐ RStudio opens without an error.
- ☐ The Console shows an R version.
- ☐ 1 + 1 returns 2.
- ☐ You can create an R Script and run code from it.

Please check this before the course: Completing the installation in advance will allow us to start the hands-on R exercises directly during the Training School.

Official sources

If version numbers or system requirements change, always follow the latest information on these official pages:

CRAN - The Comprehensive R Archive Network: <https://cran.r-project.org/>

CRAN - R for macOS: <https://cran.r-project.org/bin/macosx/>

Posit - RStudio downloads: <https://posit.co/downloads/>

Posit - RStudio IDE User Guide: <https://docs.posit.co/ide/user/>